

AI-Powered Credit Risk Intelligence Platform

Explainable default scoring · Talk-to-Data · Decision rules



Problem & Objective

Banks need faster, more accurate, EXPLAINABLE credit decisions.

- Identify high-risk applicants early and automate risk scoring
- Provide explainable reasons for every classification (audit/regulatory)
- Let analysts explore data in plain English
- Bridge ML insights and credit policy with business rules

One Dockerized app covering EDA → ML → XAI → Rules → Chatbot.

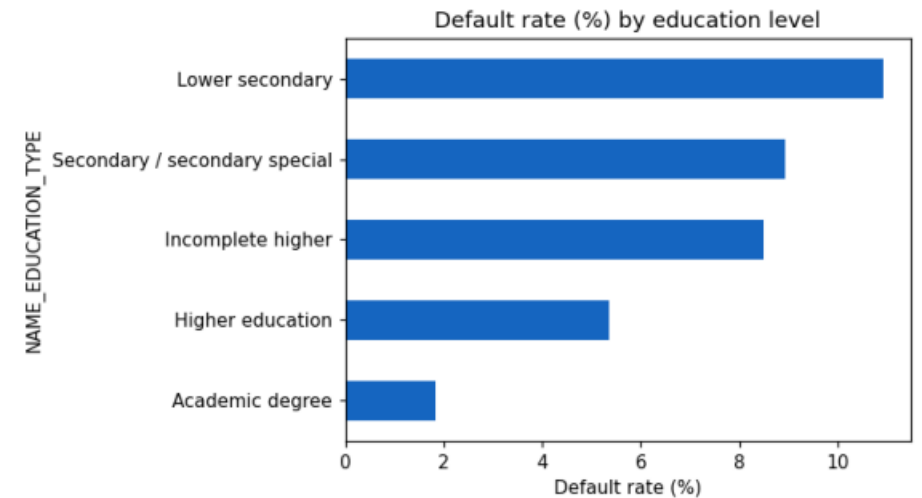
Architecture

Single FastAPI container serves a JSON API + static frontend.

- Frontend (HTML/JS/Chart.js): 5 tabs
- /api/predict → LightGBM model (model.pkl)
- /api/explain → SHAP TreeExplainer
- /api/rules → decision-tree rules (rules.json)
- /api/chat → Gemini NL→SQL + SQL safety layer → SQLite
- Shared preprocessing: identical at train & predict time
- bootstrap.py builds artifacts on first run → one-command deploy

EDA — Key Findings

- Only 8.07% default → heavy class imbalance
- Lower education defaults ~2x higher than higher education
- Younger applicants (20–30) default markedly more
- External scores (EXT_SOURCE_*) are the strongest signals
- Low-skill Laborers / Drivers: highest occupational default (>13%)
- DAYS_EMPLOYED has an 18% sentinel anomaly (pensioners)



Default rate (%) by education level

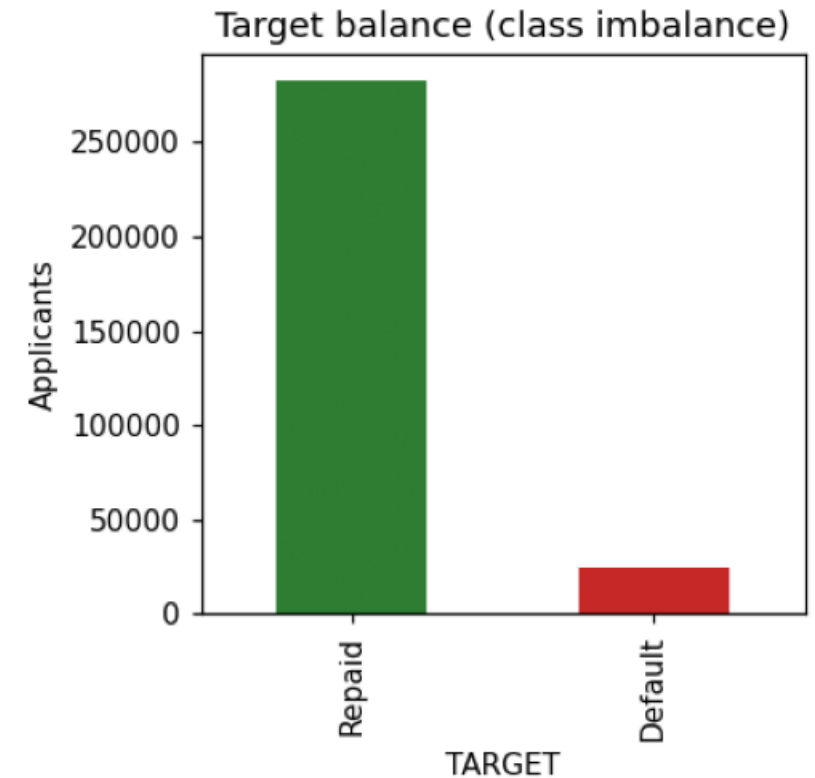
Class Imbalance — A Data-Driven Choice

Tested `scale_pos_weight` & `is_unbalance` — BOTH hurt:

- Degenerate early-stopping; ROC-AUC fell $0.77 \rightarrow 0.72$

Our strategy instead:

- Keep natural distribution (LightGBM `BoostFromScore` handles base rate)
- Judge with ROC-AUC & PR-AUC, never plain accuracy
- Tune decision threshold to maximize F1 (not 0.5)



8% default — class imbalance

Model Performance (held-out 20%)

0.770

ROC-AUC

0.259

PR-AUC

0.738

Baseline AUC

0.148

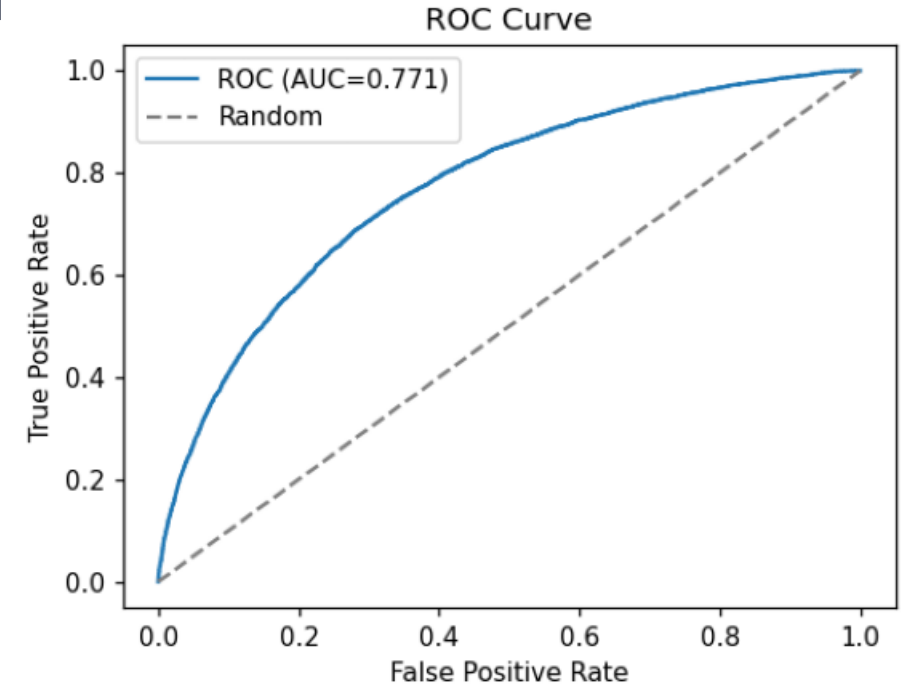
F1 threshold

LightGBM beats the Logistic-Regression baseline. PR-AUC is the honest metric under 8% imbalance. Risk bands: Low <0.084 , Medium <0.182 , High ≥ 0.182 .

Evaluation Curves

ROC-AUC measures ranking quality across all threshold

- PR-AUC (right) is more honest under heavy imbalance.
- Base rate ~8% shown as the dashed line.

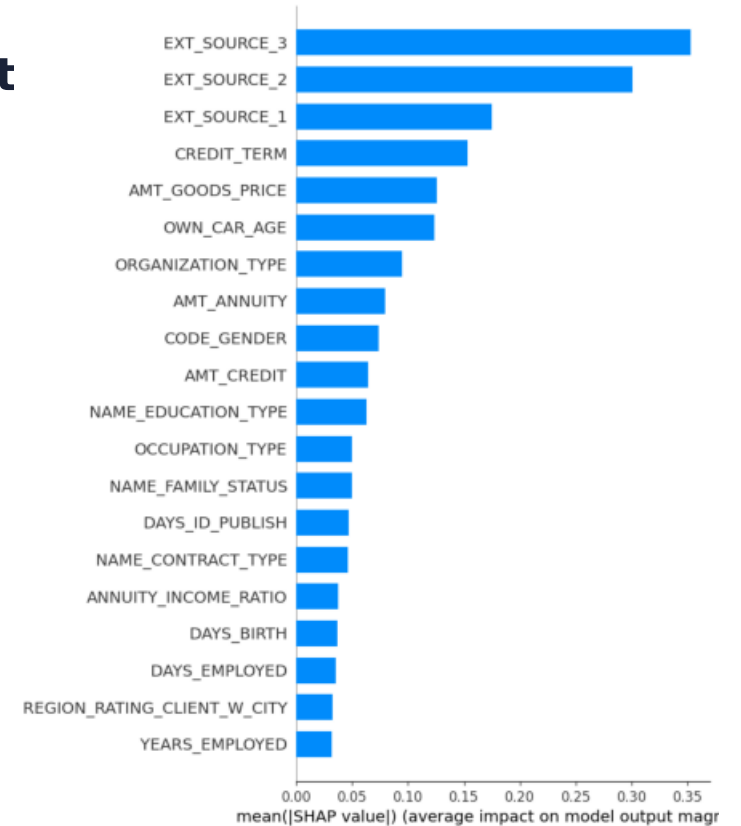


ROC curve (AUC = 0.770)

Explainable AI (SHAP)

Every prediction is decomposed into per-feature contribution

- TreeExplainer — exact & fast on LightGBM
- UI shows top drivers pushing risk UP (red) and DOWN (green)
- Plain-English phrases, e.g. 'Low EXT_SOURCE_2 increased risk'
- Satisfies audit / regulatory explainability needs



Global SHAP importance

Business Decision Rules

Shallow decision tree → readable IF/THEN credit rules.

- Each rule: support, observed default rate, lift vs 8% base

Sample:

- $\text{EXT_SOURCE_3} \leq 0.32$ AND $\text{EXT_SOURCE_2} \leq 0.08 \rightarrow 37.6\%$ default (4.66x, n=2079)
- $\text{EXT_SOURCE_3} \leq 0.32$ AND $\text{EXT_SOURCE_3} \leq 0.15 \rightarrow 31.4\%$ default (3.89x, n=2845)

Bridges ML scores and human credit policy.

Talk-to-Data — NL → SQL

Natural-language questions → SQL → readable answers (Gemini).

Prompt & token optimization:

- Curated 24-col schema (not all 122), cached; few-shot; temp=0

Hallucination / safety control (independent layer):

- SELECT-only, single statement, known table only, enforced LIMIT
- Read-only SQLite connection; UI shows exact SQL + rows (auditable)

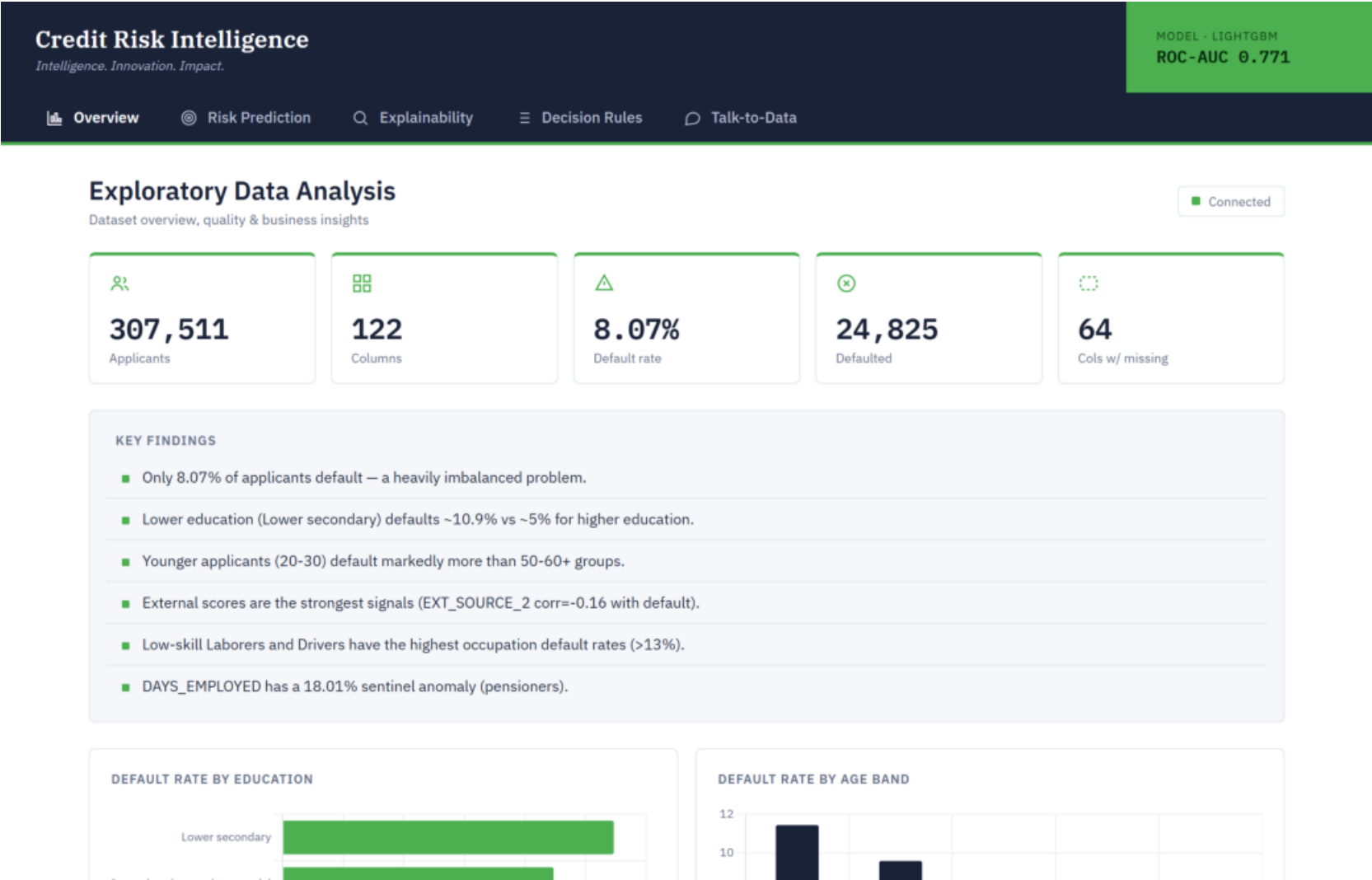
Deployment & Deliverables

- `docker-compose up --build` → `http://localhost:8000`
- Dockerfile + `docker-compose.yml` + `.env.example`
- Dataset mounted (not baked); artifacts persisted
- bootstrap builds DB/model/EDA/rules if missing

Limitations & next steps:

- Single-table model — add bureau/previous aggregates → AUC ~0.78
- Hyperparameter tuning; probability calibration

Application — Overview / EDA



Dataset summary, data-quality KPIs and business-insight charts.

Application — Risk Prediction

Annual income	Loan amount	Loan annuity
150000	600000	27000
Goods price	Age (years)	Years employed
540000	35	5
Children	External score 1	External score 2
0	0.5	0.5
External score 3	Gender	Contract type
0.5	M	Cash loans
Education	Owns car	
Secondary / secondary special	N	

Assess Risk



4.3%
DEFAULT PROB.

LOW RISK

4.3% Default probability
0.771 Model ROC-AUC
0.259 Model PR-AUC

Applicant form → default probability, risk band and live model metrics.

Application — Explainability (SHAP)

Credit Risk Intelligence
Intelligence. Innovation. Impact.

MODEL - LIGHTGBM
ROC-AUC 0.771

[Overview](#) [Risk Prediction](#) [Explainability](#) [Decision Rules](#) [Talk-to-Data](#)

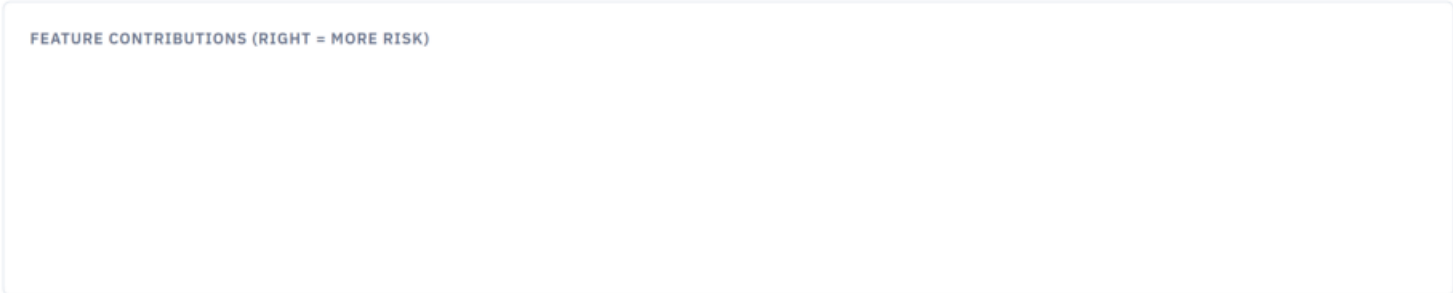
Explainability

Why the model made this decision (SHAP)

Run a prediction first — this explains that same applicant with SHAP.

Explain last applicant

FEATURE CONTRIBUTIONS (RIGHT = MORE RISK)



 Computing SHAP values...

Per-applicant risk drivers: red increases risk, green decreases it.

Application — Decision Rules

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Decision Rules

Business-readable credit rules derived from ML

Connected

Population base default rate: 8.1%. Rules sorted by deviation from base.

RISK	RULE	DEFAULT RATE	APPLICANTS	LIFT
HIGH	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 <= 0.380 AND EXT_SOURCE_2 <= 0.080	37.6%	2,079	4.66×
HIGH	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 <= 0.380 AND EXT_SOURCE_2 > 0.080 AND EXT_SOURCE_3 <= 0.146	31.4%	2,845	3.89×
HIGH	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 > 0.380 AND EXT_SOURCE_3 <= 0.140 AND EXT_SOURCE_2 <= 0.633	22.9%	4,405	2.84×
HIGH	EXT_SOURCE_3 > 0.315 AND EXT_SOURCE_2 <= 0.391 AND EXT_SOURCE_3 <= 0.542 AND EXT_SOURCE_2 <= 0.153	21.7%	9,024	2.69×
HIGH	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 <= 0.380 AND EXT_SOURCE_2 > 0.080 AND EXT_SOURCE_3 > 0.146	21.2%	7,893	2.63×
MEDIUM	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 > 0.380 AND EXT_SOURCE_3 <= 0.140 AND EXT_SOURCE_2 > 0.633	13.9%	2,542	1.72×
LOW	EXT_SOURCE_3 > 0.315 AND EXT_SOURCE_2 > 0.391 AND EXT_SOURCE_3 > 0.536 AND EXT_SOURCE_2 > 0.625	2.4%	49,031	0.3×
MEDIUM	EXT_SOURCE_3 > 0.315 AND EXT_SOURCE_2 <= 0.391 AND EXT_SOURCE_3 > 0.542 AND EXT_SOURCE_2 <= 0.131	12.8%	4,145	1.59×
MEDIUM	EXT_SOURCE_3 <= 0.315 AND EXT_SOURCE_2 > 0.380 AND EXT_SOURCE_3 > 0.140 AND EXT_SOURCE_1 <= 0.527	12.8%	19,840	1.59×

Business-readable IF/THEN rules with default rate, support and lift.

Application — Talk-to-Data Chatbot

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Talk to the Data

Ask questions in plain English

Connected

What is the overall default rate?

Show the default rate by education level, highest first.

What is the average income of defaulters versus non-defaulters?

Which 5 occupation types have the highest default rate (min 1000 applicants)?

How many applicants own a car, and what is their default rate?

Compare default rate for cash loans versus revolving loans.

What is the overall default rate?

Ask

The overall default rate across all applications is 8.07%.

GENERATED BY GEMINI - PROMPT V1

```
SELECT ROUND(AVG(TARGET) * 100, 2) AS default_rate_pct FROM applications LIMIT 100
```

DEFAULT_RATE_PCT
8.07

Natural-language question → generated SQL → readable answer + table.

Thank you

End-to-end, explainable, and reproducible.



Code, README, and artifacts in the repository.